How Reliable Is One Kernel SHAP Run? Budget-Conditional Top-3 Reproducibility and a Single-Run Self-Diagnostic

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Abstract

Explanations of tabular predictions are often produced by one run of a stochastic Shapley estimator, and the resulting top features are then reported as the reason for the prediction. We ask when the top-3 features of one budget-limited Kernel SHAP run would be reproduced by independent reruns. We define reproducibility as the mean top-3 Jaccard agreement between the assessed run and eight reruns, and we study coalition budgets of $4d$ and $16d$ for d features. On three held-out tabular datasets and three model families, mean agreement is 0.807 at $4d$ and 0.932 at $16d$, so single-run reliability depends on the budget. We then test whether the assessed run's own coalition data can predict its reproducibility, using a jackknife-based rerun simulation shrunk toward a budget prior. In the sealed evaluation, which qualified, the predictor lowers squared error against always-reliable, constant-0.5 and split-half-gap baselines, with positive one-sided bootstrap lower bounds, and beats a pairing-permuted control by 0.0137. The permuted control keeps about 61% of the gain over always-reliable, per-instance rank correlation is 0.58, and constants are hand-set and untuned. The results are limited to Kernel SHAP, three datasets and two budgets.

1 Introduction

Shapley-value attributions (Lundberg & Lee, 2017) are used to explain individual predictions of tabular models in credit, medical and similar settings, and practitioner studies report feature attributions as a widely deployed form of explanation (Bhatt et al., 2020). For models without a tractable exact algorithm, they are usually estimated by a stochastic procedure under a fixed budget of model evaluations. Two runs of the same estimator on the same trained model and the same instance therefore return different attribution vectors, and the features that appear in a short list of reasons can change from run to run. A common practice, which we take as the case of interest, is to run the explainer once and report what it returns. Whether that practice is adequate, and under which conditions, is rarely stated as a checkable claim.

We study a deliberately narrow version of this question. The explainer is Kernel SHAP with an interventional value function and a 32-row background, run with a budget of $B = 4d$ or $B = 16d$ coalitions for a model with d features. The decision-relevant object is the top-3 feature set by absolute attribution. We call a single run *reproducible* to the degree that independent reruns, which differ only in their coalition sampling, return the same top-3 set. This definition ignores background resampling, model retraining and the choice between explainers; those sources of disagreement are outside our scope and we do not measure them.

The study has two parts. First, we measure how reproducible single runs are at the two budgets, on three tabular datasets and three model families that were held out from the selection of the diagnostic. Second, we ask whether reproducibility can be *predicted for a given run from that run alone*. A practitioner holds one run, not eight. A self-diagnostic that reads the run's coalition masks, weights and evaluated coalition values and returns a reproducibility estimate $\hat{r} \in [0, 1]$ would let the practitioner flag instances whose top-3 set should not be trusted. We evaluate such a diagnostic against three baselines and against a control that keeps all episode-level information but destroys instance-level pairing.

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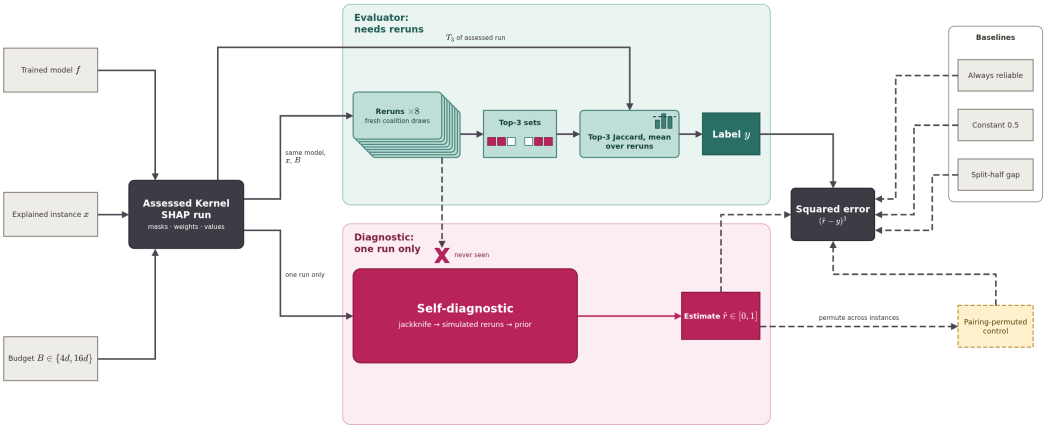


Figure 1: Protocol overview. A trained model and an explained instance yield one assessed Kernel SHAP run at budget $4d$ or $16d$. Eight independent reruns define the reproducibility label (mean top-3 Jaccard with the assessed run). The self-diagnostic sees only the assessed run and returns $\hat{r}$, which is scored by squared error against the label and compared with three baselines and a pairing-permuted control.

Contributions.

- An operational, rerun-based reproducibility label for single-run top-3 attribution, together with a squared-error evaluation protocol that includes trivial baselines, a split-half-gap heuristic and a pairing-permuted control (Section 3).
- A measurement of budget dependence. Across the held-out datasets and model families, mean top-3 Jaccard agreement is 0.807 at $4d$ and 0.932 at $16d$ (Section 4).
- A single-run diagnostic that combines a delete-one-group jackknife of the run's own regression with Monte-Carlo propagation to the top-3 set and shrinkage toward a budget prior. It beats every baseline and the permuted control with positive lower bounds, but the margin over the control is small in absolute terms (0.0137), and most of its gain over the always-reliable baseline survives permutation.
- An explicit account of what is not established: no ablation of the diagnostic's components, no per-dataset or per-family breakdown in the retained evidence, only two budgets, and constants set by hand (Section 5).

2 Related Work

Shapley estimation. Kernel SHAP casts Shapley estimation as a weighted regression over sampled coalitions (Lundberg & Lee, 2017). Later work reduces its variance through paired sampling and regression refinements (Covert & Lee, 2020), permutation sampling (Mitchell et al., 2021), leverage-score sampling with error guarantees (Musco & Witter, 2024), and amortized estimators (Jethani et al., 2021). Gupte & Paparrizos (2025) compare Shapley approximations empirically on tabular data. These papers make the estimator more accurate or give accuracy guarantees at a chosen budget. We take the budget as given and ask what a user holding one run can say about the reproducibility of its top-3 set, which is a statement about the run rather than about the estimator's average error. The closest work on sampled local explainers asks the per-explanation version of this question for LIME: Visani et al. (2021) define stability indices from repeated runs, and Slack et al. (2020) model the uncertainty of LIME and Kernel SHAP explanations with Bayesian local regressions. Both ask for more runs or a posterior over them. We instead ask whether the top-3 set returned by one run at a fixed budget will be reproduced by a rerun, using only that run's own coalition data and an evaluator that does reruns to supply the label.

What a Shapley value means. The value function defines what is being attributed: interventional or conditional expectations answer different questions (Sundararajan & Najmi, 2019; Chen et al.,

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2020; Frye et al., 2020; Kumar et al., 2020; Covert et al., 2020), and additive attributions can misrepresent models with interactions (Gosiewska & Biecek, 2019). We fix an interventional value function with a 32-row background and treat the resulting attribution, not its causal or data-faithful interpretation, as the object whose run-to-run stability is measured.

Instability and disagreement of explanations. Local explanations are known to vary under perturbation and between methods (Ribeiro et al., 2016; Alvarez-Melis & Jaakkola, 2018; Krishna et al., 2022), and Yuan et al. (2022) study how background size affects SHAP stability for deep models. Our label isolates a different and narrower source of variation, namely the coalition sampling of a single estimator with everything else fixed. This is analogous in spirit to predictive multiplicity, which asks how much predictions vary across equally good alternatives (Marx et al., 2019), except that the alternatives here are reruns of an estimator.

Stability and similarity measures. Feature-selection stability is commonly measured by average pairwise similarity of selected sets, with corrections for chance agreement (Nogueira et al., 2017), and rank-based comparisons of incomplete lists use measures such as rank-biased overlap (Webber et al., 2010). We use the plain Jaccard index on top-3 sets because three reasons is a natural unit for a decision explanation. We do not apply a chance correction, which matters when datasets with $d = 10$ and $d = 20$ are averaged (Section 5). Uncertainty quantification for Shapley-based importance has been developed at the population level (Williamson & Feng, 2020); we address the per-instance, per-run level.

Tabular data and models. Our held-out datasets include an ACS income task from the folktables toolkit (Ding et al., 2021). Our model families are logistic regression, random forest and a small MLP implemented in scikit-learn (Pedregosa et al., 2012). Deep tabular architectures (Gorishniy et al., 2021) are not part of this study, and exact tree-specific Shapley algorithms (Lundberg et al., 2018) are not used. Our framing of reproducibility as a measurable property with explicit scope follows the call for rigorous evaluation of interpretability (Doshi-Velez & Kim, 2017).

3 METHOD

3.1 PROBLEM SETUP

Let f be a trained binary classifier and $f_1(x)$ its predicted probability of the positive class. Raw features are grouped so that a categorical feature, after one-hot encoding, forms one group; there are d groups. For an explained instance x and a background set of $m = 32$ training rows, the interventional value of a coalition $S \subseteq [d]$ is

$$v(S) = \frac{1}{m} \sum_{b=1}^{m} f_1\big(x_S,\, b_{\bar{S}}\big), \tag{1}$$

with anchors $v_0 = v(\emptyset)$ and $v_1 = v([d]) = f_1(x)$. A Kernel SHAP *run* draws B coalitions with masks $z_i \in \{0,1\}^d$ and weights w_i (summing to one), evaluates $v_i = v(S_i)$, and returns

$$\phi = \arg\min_{\phi} \sum_{i=1}^{B} w_i \big(v_i - v_0 - z_i^\top \phi\big)^2 \quad \text{s.t.} \quad \mathbf{1}^\top \phi = v_1 - v_0. \tag{2}$$

The coalition set follows the Kernel SHAP size distribution with complement pairs placed adjacently. Size classes are enumerated exhaustively when the budget affords it and the remainder is sampled in complement pairs; the weights of sampled coalitions are uniform over the remaining mass. The budget is $B = c\,d$ with $c \in \{4, 16\}$, which gives $B \in \{40, 160\}$ for $d = 10$ and $B \in \{80, 320\}$ for $d = 20$ (evaluator source: `source_evidence/evaluator_final.py`, `kernel_coalitions`, lines 761–812).

Let $\mathrm{T}_3(\phi)$ be the indices of the three largest $|\phi_j|$, and let $\mathrm{J}(A, B) = |A \cap B|/|A \cup B|$. For two top-3 sets with overlap $c \in \{0, 1, 2, 3\}$ the Jaccard index is $c/(6 - c)$, that is 0, 0.2, 0.5 or 1.

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Reproducibility label. For an assessed run $\phi^{(0)}$ and $R = 8$ reruns $\phi^{(1)}, \ldots, \phi^{(R)}$ that use fresh coalition draws but the same model, instance, background and budget,

$$y = \frac{1}{R} \sum_{r=1}^{R} \mathrm{J}\big(\mathrm{T}_3(\phi^{(0)}),\ \mathrm{T}_3(\phi^{(r)})\big) \in [0, 1]. \tag{3}$$

The label is conditional on the particular assessed run, so it can in principle be predicted from that run's data. It measures coalition-sampling noise only. Because it uses $R = 8$ reruns it is itself a noisy estimate of the conditional expectation, and we cannot quantify that noise from the retained evidence.

Diagnostic task. The diagnostic receives, for each of up to 20 explained instances of a trained model, the assessed run's masks, weights, evaluated values and anchors, plus d, B, $k = 3$ and the model-family name. It returns $\hat{r} \in [0, 1]$ per instance. It never sees labels, reruns, model internals or dataset names.

3.2 Design rationale

Kernel SHAP is a weighted regression, so run-to-run variation of ϕ is sampling variation of regression coefficients. The top-3 set changes between runs when this variation is large relative to the gap between the third and fourth largest $|\phi_j|$. A single run contains information about its own sampling variation: deleting random subsets of its coalition pairs and re-solving the regression shows how sensitive ϕ is to the particular coalitions drawn. Propagating that sensitivity to the top-3 set by simulation gives an estimate of the expected overlap with a rerun, which is what the label measures. Two weaknesses motivate the remaining components. The jackknife is computed from only $B/2$ coalition pairs and is itself noisy, particularly at $4d$; and its spread need not match the rerun spread. We therefore shrink the simulated value toward a budget-specific prior. We hypothesized that this would reduce error most where the label is high and nearly constant (large budget) and that instance-specific information would matter most near ties between the third and fourth feature. We did not run component ablations, so these are design motivations and not tested mechanisms.

3.3 Compared predictors

Self-diagnostic (frozen selected configuration). For each instance, with $F = 8$ folds and $P = 4$ independent random partitions:

1. Assign complement pairs of coalitions at random to folds, drop one fold at a time, and resolve the regression to obtain leave-fold-out solutions $\phi^{(-f)}$. Folds that leave no more than d rows are skipped. Define centered, scaled deviations $\Delta_f = \sqrt{(F' - 1)/F'}\,(\phi^{(-f)} - \bar{\phi}^{(\cdot)})$ over the F' valid folds, so that $\sum_f \Delta_f \Delta_f^\top$ is the delete-one-group jackknife covariance (the delete-a-group jackknife of survey sampling, Kott, 2001).
2. Simulate reruns with 1000 common-random-number draws (500 antithetic pairs, seeded by the episode seed). The diagonal simulator uses $\phi + \mu\,\sigma \odot \zeta$ with $\sigma_j = (\sum_f \Delta_{fj}^2)^{1/2}$ and $\zeta \sim \mathcal{N}(0, I_d)$. The full-covariance simulator uses $\phi + \sum_f \zeta_f \Delta_f$, which preserves the efficiency constraint. Each simulated top-3 set is compared with the observed one by Jaccard index and averaged. This gives $s_{d,1}$ (diagonal, $\mu = 1$), $s_{d,0.7}$ (diagonal, $\mu = 0.7$) and $s_{f,1}$ (full covariance, $\mu = 1$).
3. Compute the margin z_m, the gap between the third and fourth largest $|\phi_j|$ divided by the jackknife standard error of that gap, averaged over partitions along with the three simulated values.
4. Combine: $s = 0.5\,s_{d,1} + 0.2\,s_{d,0.7} + 0.3\,s_{f,1}$ and

$$\hat{r} = \mathrm{clip}_{[0.02,1]}\Big(0.6\,s + 0.4\,\pi(B/d) + 0.02\tanh\big(z_\mathrm{m}/3\big)\Big), \tag{4}$$

where the prior π is linear in $\log_2(B/d)$ between 0.84 at $B = 4d$ and 0.96 at $B = 16d$.

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On any numerical failure the output falls back to $\pi(B/d)$. The source is in `source_evidence/selected_config_program.py` (constants at lines 35–46; functions `_jackknife_cov`, `_sim_jaccard`, `_margin_z`, `_empirical_bayes` and `_diagnose_one`). The final record stores only a tree hash of the evaluated sources, so the above describes the source file of the configuration designated as selected in the search record, not a hash-verified copy.

Provenance of the constants matters for interpretation. The weights $0.5/0.2/0.3$, 0.6, 0.02, the fold count and the partition count are hand-set and were not tuned on instance labels. The values 0.84 and 0.96 are described in the configuration's notes as aggregate label means by budget from earlier search-plane evaluations (`selected_config_NOTES.md`). The shrinkage weight is therefore a fixed constant, not an estimated empirical-Bayes weight; we use the term *empirical-Bayes-style* only for the structure of shrinking an unreliable per-unit estimate toward a prior derived from other data, as in Stein-type estimators (Efron & Morris, 1973).

Baselines and control. All are scored with the same per-instance squared error $(\hat{r}_i - y_i)^2$. *Always reliable* outputs $\hat{r} = 1$. *Constant 0.5* outputs 0.5. The *split-half gap* heuristic is computed by the evaluator from the assessed run alone: it solves the regression separately on the two halves of the complement-pair blocks, sets $se_j = |\phi_j^A - \phi_j^B|/2$, forms $z = (|\phi|_{(3)} - |\phi|_{(4)})/\sqrt{se_{(3)}^2 + se_{(4)}^2}$, and outputs $0.5 + 0.5 \tanh z$. The *pairing-permuted control* scores the diagnostic's own outputs against labels after permuting them among the 20 instances of an episode with each of 20 fixed derangements and averaging. It therefore keeps everything the diagnostic knows about the episode, such as its budget, model and dataset level, and removes only the instance-level pairing between $\hat{r}$ and y. The diagnostic exceeds this control only if its instance-specific variation is informative.

3.4 Experimental protocol

The evaluation was run once under a sealed protocol, with the selected configuration frozen beforehand. Datasets are the ACS income task (California, 2018, 1-year person records via folktables; $d = 10$; 20,000 sampled rows), Give Me Some Credit (OpenML 46929; $d = 10$; 20,000 sampled rows) and the second half of a hash-based split of Statlog German Credit ($d = 20$; 485 rows). Models are scikit-learn logistic regression, a 100-tree random forest and a one-hidden-layer MLP with 64 units, trained on a 70% split with train-only median imputation and standardization. For each dataset, family, budget in $\{4d, 16d\}$ and seed in $\{9001, 9002\}$ there is one episode of 20 explained test instances, with nine runs per instance (one assessed, eight reruns). That gives 36 episodes and 720 labeled instances. Scores aggregate with equal weight over instances, then seeds within a dataset–family–budget cell, then cells within a dataset, then datasets. The headline quantity is score $= 1 -$ MSE of the diagnostic. A *skill* is the MSE of a comparator minus the MSE of the diagnostic. The evaluator reports one-sided 2.5th-percentile lower bounds from 200 hierarchical bootstrap replicates that resample cells, seeds and instances within each of the three fixed datasets. These datasets were disjoint, by normalized record hash, from the two search-plane datasets (a hash-based split of German Credit and Pima Diabetes) used to choose the configuration (`final_resources_manifest.json`, `search_resources_manifest.json`). The evaluator's pre-declared assessment treats the diagnostic as informative only if the three skill lower bounds are positive and its score exceeds the permuted control by a positive lower bound (`task_contract.json`, `protocol.promotion_rule`).

4 Results

All numbers in this section come from the sealed final evaluation unless stated otherwise. The evaluation completed and its recorded outcome is *qualified*. The pre-declared assessment rule from the protocol is met by the reported lower bounds, all four of which are positive. This is a statement about the diagnostic's measured skill in this regime, and not a validity, superiority or generality claim about Kernel SHAP reliability.

4.1 Single-run reproducibility depends on the budget

Mean top-3 Jaccard agreement is 0.8065 at $B = 4d$ and 0.9315 at $B = 16d$ (Figure 2b). The difference is 0.125. Since a one-feature swap gives Jaccard 0.5 and larger disagreements give at

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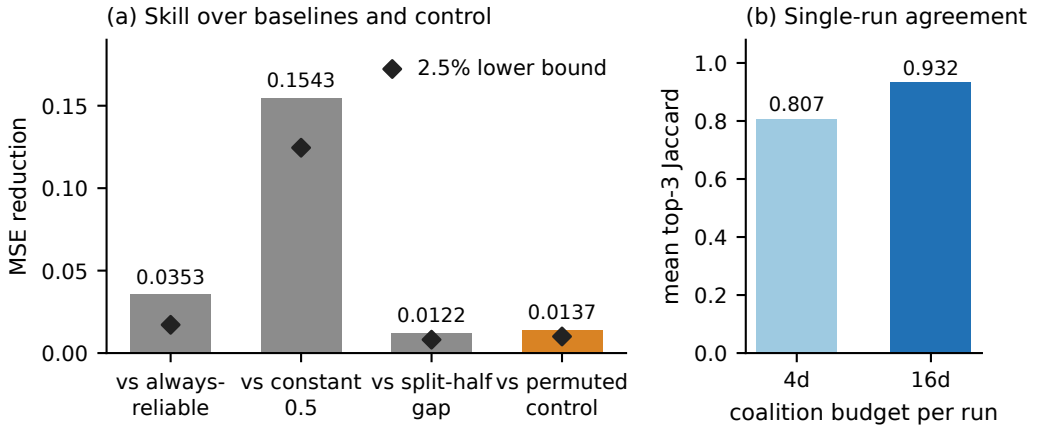


Figure 2: Sealed final evaluation. (a) Reduction in mean squared error of the single-run diagnostic relative to each baseline and to the pairing-permuted control, with the evaluator's one-sided 2.5th-percentile bootstrap lower bound (diamonds). (b) Mean top-3 Jaccard agreement between an assessed run and eight reruns at the two budgets, averaged over three held-out datasets, three model families, two seeds and 20 instances per cell.

Table 1: Sealed final evaluation of the frozen self-diagnostic. Scores are $1 - \text{MSE}$ against the rerun-agreement label; skills are MSE reductions. Lower bounds are the evaluator's one-sided 2.5th percentiles over 200 bootstrap replicates; no upper bounds were recorded. Columns marked † are derived from the reported values (`analysis/derived_metrics.json`).

Quantity	Estimate	Lower bound	MSE† / share†
Diagnostic score	0.9838	–	0.0162
Skill vs. always reliable	0.0353	0.0171	0.0515
Skill vs. constant 0.5	0.1543	0.1245	0.1705
Skill vs. split-half gap	0.0122	0.0081	0.0284
Permuted-control score	0.9701	–	0.0299
Diagnostic − control score	0.0137†	0.0101	45.9% of control MSE
Spearman $\rho(\hat{r}, y)$	0.5809	–	–
Mean label at $4d$ / $16d$	0.8065 / 0.9315	–	–
Score, seed 9001 / 9002	0.9850 / 0.9826	–	–

most 0.2, the loss $1 - \bar{y}$ bounds the share of assessed–rerun pairs with non-identical top-3 sets. That share is at least $1 - \bar{y}$ and at most $2(1 - \bar{y})$: between 19.3% and 38.7% at $4d$, and between 6.8% and 13.7% at $16d$ (derived; `analysis/derived_metrics.json`). Even at $16d$, then, a noticeable minority of rerun pairs disagree on the identity of the top three features, and at $4d$ a substantial minority do. These are averages over datasets, model families, seeds and instances. The retained final evidence contains only per-seed aggregate scores, not per-cell labels, so we cannot say which datasets, families or instances drive them.

4.2 The diagnostic against baselines and a control

The diagnostic reaches a score of 0.9838 (MSE 0.0162, RMSE 0.127; derived). Table 1 gives the comparison. Its MSE is lower than that of always-reliable by 0.0353, of constant 0.5 by 0.1543, and of the split-half-gap heuristic by 0.0122. The lower bounds of these three skills are 0.0171, 0.1245 and 0.0081, all positive. Relative to each comparator's MSE, the reductions are 68.5%, 90.5% and 42.9% (derived). The *always-reliable* baseline is the practitioner's default of trusting every run; the mean label deficit $1 - y$, averaged over the two budgets, is only 0.131 (derived), so the room for improvement by prediction is small in absolute terms, and the split-half-gap heuristic already captures part of it.

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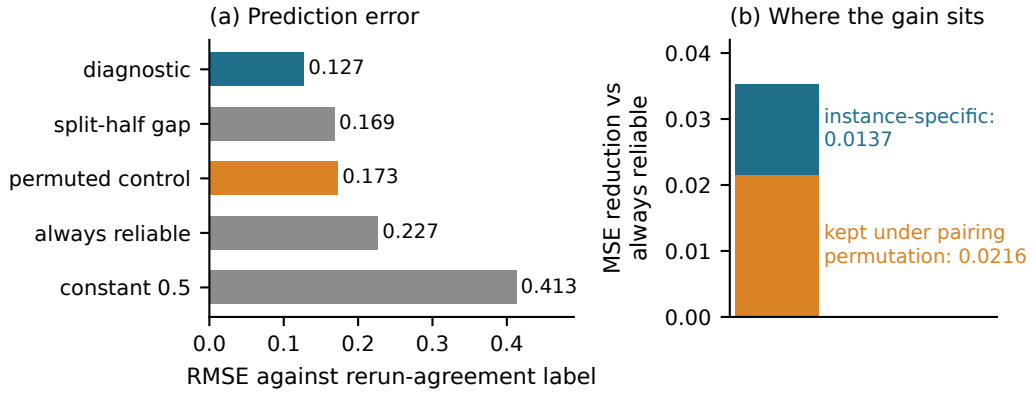


Figure 3: Error accounting derived from the sealed metrics. (a) Root-mean-squared error of each predictor against the rerun-agreement label. (b) Decomposition of the diagnostic's MSE reduction relative to always-reliable into the part that survives instance-level permutation and the instance-specific part.

4.3 How much is instance-specific?

The pairing-permuted control scores 0.9701 against 0.9838 for the diagnostic, a difference of 0.0137 with lower bound 0.0101. In relative terms this is a 45.9% reduction in MSE, but in absolute terms it is small, and Figure 3b shows why this matters for interpretation. Of the 0.0353 reduction over always-reliable, 0.0216 (61%) is retained when the diagnostic's outputs are permuted within episodes, and 0.0137 (39%) is lost. The retained part is knowledge the diagnostic holds about the episode: chiefly the budget, through the prior, and any dataset- or family-level shifts in its simulated values. The lost part is the instance-specific information. Both parts are measured, and the lower bound for the instance-specific part is positive, which implies that labels vary across instances in a way that a single run can partly anticipate. The control does not separate budget calibration from dataset- or family-level calibration; the evidence has no breakdown that would allow this.

Rank resolution is moderate. The mean within-episode Spearman correlation between $\hat{r}$ and y is 0.5809. The evaluator assigns zero to episodes in which either vector is constant, which can happen when all 20 labels are 1.0; this convention may lower the average. A rank correlation of 0.58 means that $\hat{r}$ orders instances better than chance but not sharply: a threshold on $\hat{r}$ would misclassify many instances near the threshold. We did not evaluate any thresholded flagging rule, calibration curve or selective-prediction metric, so we make no claim about how well $\hat{r}$ would work as an abstention signal.

4.4 Consistency across seeds and search-plane checks

Per-seed scores are 0.9850 for seed 9001 and 0.9826 for seed 9002, a difference of 0.0025. With two seeds this gives no variance estimate, and the bootstrap lower bounds are the only recorded uncertainty. On the search plane (German Credit half A and Pima Diabetes), the selected configuration scored 0.9854, against 0.9838 on the sealed final data; this search score is a development proxy, was used for selection among three configurations, and is not an independent estimate. The prior constants 0.84 and 0.96 exceed the final label means by 0.0335 at $4d$ and 0.0285 at $16d$ (derived), so the prior was somewhat optimistic on the held-out data, and the diagnostic still qualified. How much of the final score depends on this prior is unknown, because no variant without the prior was evaluated.

5 Discussion and Limitations

What the evidence supports. Within the evaluated regime (Kernel SHAP, 32-row interventional background, $k = 3$, budgets $4d$ and $16d$, three datasets, three model families), the answer to "is one run enough?" is conditional. The mean agreement is 0.81 at the smaller budget and 0.93 at the

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larger one, and a run-internal diagnostic can predict agreement with lower error than the baselines. We do not claim that SHAP is reliable or unreliable in general, nor that any specific agreement level is sufficient for a particular use. Sufficiency depends on the cost of a top-3 flip in the application, which the label does not encode.

What the control tells us. The margin over the permuted control is positive and its lower bound is positive, but 61% of the diagnostic's gain over always-reliable survives permutation. Most of the measured benefit is therefore consistent with knowing the budget and similar episode-level facts. The instance-specific contribution is real in the sense of the bootstrap, but small in absolute MSE. A fair comparison would also include a budget-conditional constant, which is the prior alone and was not evaluated; without it, we cannot say how much the simulation adds to the prior.

No component attribution. The diagnostic combines a partition ensemble, three simulators, a prior and a margin term. There are no ablations, so the results support the claim that the assembled configuration beats the baselines and the control in this regime, and not that any component is responsible. Three configurations were scored on the search plane (Supplementary Table 5); their per-seed ranges overlap for the two best, and we draw no mechanistic conclusion from the differences.

Scope of the label. The label captures only coalition-sampling noise. Background resampling (Yuan et al., 2022), retraining noise and disagreement between explainers (Krishna et al., 2022) are not measured and would add to run-to-run variation in practice, so our reproducibility estimates are optimistic for a deployment where those vary. The label uses $R = 8$ reruns, which makes it noisy; the diagnostic's measured MSE includes this label noise, so the true prediction error of $\hat{r}$ for the conditional expectation could be smaller, and we cannot say by how much. Jaccard on top-3 sets is not corrected for chance agreement, whose level differs between $d = 10$ and $d = 20$ (Nogueira et al., 2017), and equal-weight dataset averaging mixes these regimes. Other values of k, rank-aware measures such as RBO (Webber et al., 2010), and top-1 agreement were not studied.

Statistical limits. Only two seeds and three datasets were used. The bootstrap resamples cells, seeds and instances within the fixed datasets, not the datasets themselves, so the lower bounds do not license claims about other datasets. Only one-sided lower bounds were recorded, so we do not report two-sided intervals. The permuted control uses a fixed set of 20 derangements. Search-plane scores informed the choice of the configuration and the prior constants; the final datasets are disjoint from the search datasets by record hash, but they share model families, budgets and evaluation code with the search plane.

Not done. We did not evaluate other explainers, deep tabular models, other budgets, background-size effects, retraining noise, or per-dataset, per-family and per-budget breakdowns of the diagnostic's error. Candidate-side diagnostics in the evaluator record are marked unverified, and we rely only on the evaluator's own metrics. Implementation details are taken from the readable source file; the final record identifies the evaluated sources by tree hash only.

6 Conclusion

For one budgeted Kernel SHAP run on a tabular classifier, reproducibility of the top-3 feature set is conditional on the coalition budget: mean agreement with reruns is 0.807 at $4d$ and 0.932 at $16d$ across three held-out datasets and three model families. A diagnostic that reads only the assessed run lowers squared error against always-reliable, constant-0.5 and split-half-gap baselines and against a pairing-permuted control, with positive bootstrap lower bounds, and the sealed evaluation qualified. The conclusion is bounded. The instance-specific gain over the control is 0.0137, most of the gain over the always-reliable baseline survives permutation, rank resolution is moderate (0.58), and neither the contribution of individual components nor results for individual datasets or families were established. The most useful follow-ups follow directly from these gaps: evaluating a budget-only constant and component ablations, reporting per-cell results with two-sided intervals, adding background and retraining variation to the label, and testing a thresholded flagging rule against a stated cost of top-3 errors.

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A Supplementary Information

A.1 Evaluation protocol details

Table 2 lists the datasets, and the remaining protocol constants follow. All values are read from the evaluator source and manifests in `source_evidence/`. The ACS sample is the 20,000 rows

Table 2: Datasets. Rows are after the evaluator's de-duplication and sampling. d is the number of feature groups; budgets are $4d$ and $16d$.

Plane	Dataset	Rows	d	Source
Search	German Credit, half A	515	–	UCI Statlog, hash-parity split
Search	Pima Diabetes	768	–	OpenML 37, v1
Final	ACS income (CA, 2018)	20,000	10	folktables 0.0.12, ACSIncome
Final	Give Me Some Credit	20,000	10	OpenML 46929, v1
Final	German Credit, half B	485	20	UCI Statlog, hash-parity split

with the smallest record hash after removing 5,711 duplicates from 195,665 records, and the Give Me Some Credit sample is selected the same way after removing 609 duplicates from 150,000 records. German Credit has no duplicates, and its two halves are assigned by the parity of the record hash. Per-seed sweeps (18 cells each) took 53.3 s and 53.8 s on CPU. The evaluator environment is pinned to numpy 2.5.4, pandas 3.1.1, scikit-learn 1.9.1 and scipy 1.18.1, with 8 CPU cores, 16 GB of memory and a 4500 s wall-clock limit for the final plane (`task_contract.json`). Final seeds are 9001 and 9002 and search seeds are 101 and 102. Each episode explains 20 test instances, uses 8 reruns, and the control uses 20 deterministic derangements.

A.2 Derived quantities

Table 3 lists every quantity derived from the sealed metrics. With score $= 1 -$ MSE, the MSE of a comparator is one minus the diagnostic score plus its skill, and the permuted-control MSE is one minus the control score. All values are produced by `analysis/derive_metrics.py` from `paper_data.json`. The bounds on non-identical top-3 sets follow from the Jaccard values

Table 3: Derived error quantities. Reductions are relative to each comparator's MSE.

Predictor	MSE	RMSE	MSE reduction by diagnostic
Diagnostic	0.0162	0.127	–
Split-half gap	0.0284	0.169	42.9%
Permuted control	0.0299	0.173	45.9%
Always reliable	0.0515	0.227	68.5%
Constant 0.5	0.1705	0.413	90.5%

$\{0, 0.2, 0.5, 1\}$ for overlaps $\{0, 1, 2, 3\}$: a pair with non-identical sets loses between 0.5 and 1 of the maximal agreement, so a mean loss D implies a non-identical share q with $D \le q \le 2D$. The gain over always-reliable splits into 0.0216 retained under permutation (the always-reliable MSE minus the control MSE) and 0.0137 instance-specific.

A.3 Selected configuration constants

The notes accompanying the configuration state that it was written without executing any code, so no local calibration or cross-validation exists for it and none is claimed. Its search-plane score was obtained only from the evaluator.

A.4 Search-plane scores of the three configurations

Three configurations were scored on the search plane; all three completed successfully and none was unscored. The search label is evaluator-defined, so it is the same quantity as in the final evaluation,

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Table 4: Constants of the frozen self-diagnostic. None was fitted to instance labels.

Constant	Value	Role
Folds / partitions	8 / 4	delete-one-group jackknife, averaged
Draws	1000 (500 antithetic)	common random numbers per episode
Simulator weights ($s_{d,1}$, $s_{d,0.7}$, $s_{f,1}$)	0.5, 0.2, 0.3	convex combination
Prior at $B = 4d$ / $16d$	0.84 / 0.96	from earlier aggregate label means
Prior weight	0.4	shrinkage toward prior
Margin term	0.02, scale 3	soft gate, $\tanh(z_{\mathrm{m}}/3)$
Output clip	$[0.02, 1]$	numerical safety

but the datasets differ, and these scores are development proxies. Table 5 lists them. Configuration A is a single-partition jackknife with full covariance and a $\sqrt{2}$ inflation of the simulated spread and 400 draws. Configuration B replaces this with common-random-number draws, diagonal and full-covariance simulators and a fixed convex combination. Configuration C, the selection, adds the partition ensemble, the budget prior and the margin term (`searchplane_config_A_program.py`, `searchplane_config_B_program.py`, `selected_config_program.py`). Because several changes were made at once and no ablation was run, no change is credited with a score difference. The per-seed ranges of B and C overlap, and A is lower than both on each of the two seeds.

Table 5: Search-plane scores ($1 - \mathrm{MSE}$) by seed. Search seeds are 101 and 102.

Configuration	Seed 101	Seed 102	Mean
A (inflated full-covariance jackknife)	0.9740	0.9779	0.9760
B (common-random-number combination)	0.9824	0.9849	0.9837
C (selected: B + partitions, prior, margin)	0.9840	0.9869	0.9854

A.5 Reproducibility statement

The evaluator source, task contract, manifests and the source of the selected configuration are in `source_evidence/`. The scripts `analysis/derive_metrics.py` and `analysis/build_figures.py` regenerate Table 3 and Figures 2 and 3 from `paper_data.json`. Figure 1 is a schematic with no quantitative content. We did not re-run the sealed evaluation, and per-cell labels and per-instance outputs from it are not part of the retained evidence.

A.6 LLM usage

Large language models were used in this work in two roles. First, an LLM-based research agent proposed and implemented the candidate diagnostics, and the sealed evaluator, not the agent, scored them. Second, an LLM drafted this manuscript from the recorded evaluator outputs and source files, and generated the analysis and figure scripts. Every number in the paper was checked against the recorded evaluator outputs or computed from them by the included scripts. No LLM output was used as a source of experimental results. References were retrieved from bibliographic databases; our descriptions of cited works rely on their public titles and on the model's prior knowledge, and the papers were not re-read in full while writing.

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